PROJECT PROPOSAL

3D Hybrid Renewable Energy Power Plant Simulation Using OpenGL

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Course: Computer Graphics and Image Processing	**Course No.:** CSE 4101

1. Introduction

This project will create a simple interactive 3D renewable-energy scene using OpenGL. The scene combines a small hydroelectric dam, wind turbines, solar panels, water, and a few animated objects. The main purpose is not to simulate real electrical generation or complex water physics. Instead, the project is designed to clearly demonstrate the core Computer Graphics concepts taught in class through a visually meaningful scene.

2. Project Description

The user will view a 3D power-plant environment where different renewable-energy components operate together. Two wind turbines will rotate continuously, a cloud will move across the sky, a boat will travel on the water, the dam gate will move vertically, and a hydro turbine/water wheel near the water outlet will rotate. All motion will be updated continuously inside the OpenGL render loop so the animation remains dynamic and easy to understand.



Figure 1. Concept visualization of the proposed 3D hybrid renewable-energy scene.

3. Main Animated Objects

The project will contain more than five clearly visible moving objects. Their motion is directly connected to the scene, so the animation does not look artificial or unnecessary.

Moving object	Main transformation	Planned motion
Wind Turbine 1	Rotation	Three blades rotate continuously around the hub.
Wind Turbine 2	Rotation	A second turbine rotates as an independent scene object.
Cloud	Translation	Moves slowly across the sky and loops back.
Dam Gate	Translation	Moves upward and downward to show opening and closing.
Boat	Translation + rotation	Moves across the water and can slightly change direction.
Hydro Water Wheel	Rotation	Rotates continuously beside the flowing-water outlet.

4. Computer Graphics Concepts

- Translation - movement of the cloud, boat, and dam gate.
- Rotation - wind-turbine blades and hydro water wheel.
- Scaling - construction and resizing of solar panels, towers, blades, gate parts, and other reusable 3D objects.
- Transformation composition - combining translation, rotation, and scaling to place and animate objects correctly.
- Hierarchical transformation - turbine blades rotate around a hub while the complete turbine remains positioned in the scene.
- Continuous animation - object states are updated every frame inside the render loop.

5. Lighting and Shading

The scene will use a directional light to represent sunlight. The Phong lighting model will be used with ambient, diffuse, and specular components. Specular reflection will be especially visible on the water surface, solar panels, metallic gate, and turbine parts. This will make the 3D shapes easier to understand and will demonstrate lighting as a graphics concept rather than only as decoration.

6. Scene and Main Objects

- Hydroelectric dam with movable gate and water outlet
- Two wind turbines with rotating blades
- Solar panel array in the foreground
- Reservoir/water surface and a small moving boat
- Rotating hydro turbine/water wheel near the outlet
- Moving cloud, simple ground, sky, and supporting structures

7. Tools and Implementation

The project will be developed in C/C++ using OpenGL. Simple reusable 3D primitives will keep the scene efficient and easy to explain. The program will use a continuous render loop, transformation matrices, camera control, lighting calculations, and time-based animation. Keyboard controls may be added for the camera, pausing animation, opening/closing the gate, and changing selected animation speeds.

8. Expected Outcome

The final result will be a continuously running 3D hybrid renewable-energy scene. At least six independently animated objects/components will be visible. The project will clearly demonstrate 3D modeling, transformations, continuous animation, Phong lighting, and visible specular reflection while remaining simple enough for a course project.